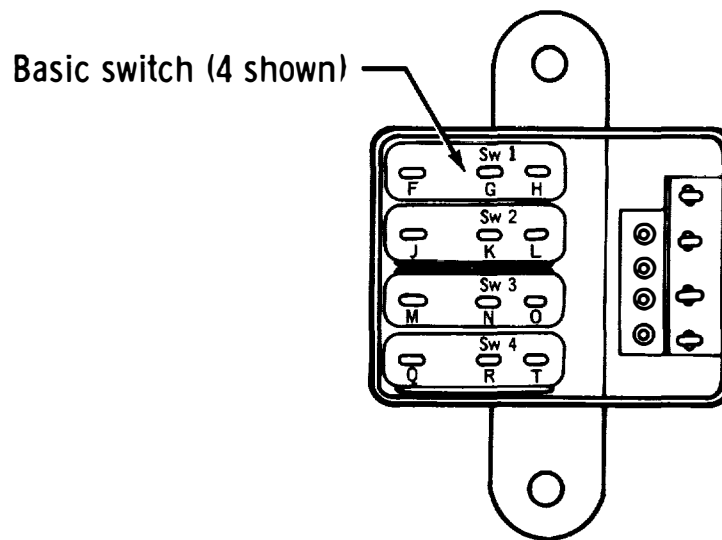




(b) Bottom view.

Figure 2. - Concluded.

TABLE II. - PUSHBUTTON SWITCH COMPARISON

Switch	Light	Number of basic switches	Movement	Function
115	EL ^a and I ^b	4	Lock and button return	Stop
116	EL	4	Lock	Abort stage
117	EL and I	2	Momentary	Start
118	I	2	Momentary	Master alarm
119	EL	4	Lock	Abort
120	EL	4	Momentary	X-translation

^aElectroluminescent lamp.

^bIncandescent lamp.

During vehicle testing, it was determined that when the engine stop switches were depressed, they could be inadvertently reset, which caused the engine to start again. To preclude this action, an external latch mechanism was added to this switch. The external latch mechanism was designed so that two actions were required to reset the stop switch, thereby eliminating the possibility of an engine restart. The design of this latch mechanism created many other problems before the latch could be properly installed in the vehicle. At present, this mechanism is working properly and has not caused any problems.

Pushbutton switches have performed well in flight. No failures have been attributed to these switches. The only recommendation as a result of this review is to make certain, during initial system design, that electronic circuits perform all complicated system switching and that mechanical switches be used to actuate this switching.

Variable Controls

Potentiometers. - Potentiometers are used within the spacecraft to provide continuous level adjustment, generally as volume controls. Each LM contains 14 of these devices in three basic configurations. The configurations include linear taper, nonlinear taper, clockwise rotation, and nonlinear taper counterclockwise rotation. The potentiometers are contained in a hermetically sealed housing that uses a bellows assembly to allow shaft rotation. Mechanical stops limit rotation to 300° . Basic resistance is provided by a carbon element that can vary between 2.5 and 25 000 ohms. Development, including qualification and acceptance testing, was satisfactory in that few failures occurred. Failures that did occur included leakage and dielectric absorption, which were easily detected and corrected.

Use of these potentiometers in the spacecraft presented numerous problems. During panel acceptance testing, many potentiometers were found to have been open or discontinuous. Analysis revealed that this problem was caused by inadequate contact pressure due to tolerance buildup between contact and contact cavity. Acceptance testing did not reveal this lack of buildup, because the potentiometers were checked only at four points. Corrective action included selective fitting of the contact into the contact housing and a screening test. The screening test consisted of monitoring the resistance over the dynamic range of the potentiometer. These potentiometers used a simple carbon element to provide resistance changes. Attempts have been made to use these devices as precision potentiometers, without success. Using these potentiometers as volume controls presented no problems during vehicle testing or flight.

Synchro. - The synchro is a rotary control used to actuate a synchro transmitter. This transmitter generates position commands to an antenna drive assembly in either the pitch or the yaw axis. The synchro is housed in a hermetically sealed case and uses a bellows assembly to provide continuous shaft rotation. Construction is that of a standard synchro with a single-phase rotor and a three-phase, delta-wound stator with carbon brushes. Development of this device was successful, resulting in only one problem. During qualification testing, the rotating (slip) torque increased after